

NEXUS AI

Product Design & Vision Document

N01–N06 • Idea & Vision • PRD • MVP Scope • Personas • Features • Product Roadmap

VERSION 0.1 • PRODUCT FOUNDATION

Product concept: NEXUS AI is an offline-first local AI workstation that orchestrates multiple local language models, retrieval-augmented knowledge, persistent project memory, controlled tools and agentic workflows through one unified application.

Core promise: One local AI environment that can chat, reason, retrieve knowledge, work with files, assist with software development and coordinate specialized local models without requiring a cloud AI subscription for its core experience.

USER → ORCHESTRATOR → MODEL / RAG / TOOL / AGENT → VERIFICATION → RESULT

N01 — NEXUS AI Idea & Vision

NEXUS AI is designed as a local-first AI operating environment rather than a single chatbot. Its purpose is to unify multiple locally available models behind an intelligent orchestration layer that can select the appropriate model, retrieve private knowledge, use controlled tools and verify results.

The first implementation will use the user’s existing Ollama ecosystem as the local model runtime. Current installed models provide an initial pool for reasoning, coding, general conversation and lightweight tasks; NEXUS should benchmark and route models rather than permanently hard-code assumptions about their quality.

Vision

Build a private AI workstation where the user's machine becomes a capable local AI environment for conversation, research, knowledge retrieval, software development, file understanding and agentic workflows.

Mission

Make advanced AI workflows available locally with privacy, transparency, model choice, controllable permissions and no per-request cloud dependency for core functionality.

Product Principles

Local-first • Privacy-first • Model-agnostic • Evidence-aware • Tool-controlled • Human-supervised • Observable • Modular • Extensible • Resource-aware.

Long-Term Vision

NEXUS should evolve from local chat into a personal AI operating layer capable of orchestrating specialist models and agents across knowledge, coding, research and controlled computer tasks.

Non-Goal

NEXUS should not claim that small local models have the same raw capability as the largest commercial frontier models. Its advantage comes from orchestration, retrieval, tools, memory, verification and privacy.

N02 — Product Requirements Document (PRD)

The PRD defines the product behavior required for the first release and the foundations needed for later agentic capabilities.

Requirement	Description	Priority
Local model access	Connect to locally installed Ollama models and expose model health/status.	P0
Unified chat	Provide persistent conversations with streaming responses.	P0
Model selection	Allow manual model selection and system-selected routing.	P0
Intelligent routing	Classify requests and select a suitable local model.	P0
RAG knowledge	Import local documents and retrieve relevant context.	P0
Citations	Show source/document references for knowledge-grounded answers.	P0
Workspace	Create projects with scoped context and files.	P0
Tool framework	Controlled interfaces for files and other approved tools.	P1
Coding agent	Inspect, edit and test code in a controlled workspace.	P1
Memory	Conversation and project memory with user controls.	P1
Verification	Evaluate generated answers/code before presenting high-impact results.	P1
Model benchmark	Compare installed models on repeatable tasks.	P1
Offline mode	Core chat, RAG and local tools work without internet.	P0
Permissions	Explicit tool and filesystem permissions.	P0
Observability	Show model, latency, token/context and tool execution status.	P1

Functional Requirements

FR-01 model discovery; FR-02 conversations; FR-03 model routing; FR-04 local knowledge retrieval; FR-05 project context; FR-06 controlled tools; FR-07 agent step records; FR-08 source attribution; FR-09 safe cancellation/recovery; FR-10 local system/model status.

Non-Functional Requirements

Privacy: core requests need no cloud transmission. Reliability: one model/tool failure must not crash the app. Performance: avoid unnecessary model loading. Security: least-privilege tools. Maintainability: replaceable adapters. Usability: advanced agents must not complicate normal chat.

N03 — Product Scope & MVP Definition

The MVP proves the central NEXUS value proposition without attempting every future agent capability.

MVP Area	Included	Excluded from MVP
Local AI	Ollama connection, model discovery, chat	Training/fine-tuning models
Chat	Conversations, streaming, history	Complex cloud collaboration
Routing	Basic task classification and model routing	Fully autonomous model marketplace
RAG	Local documents, embeddings, retrieval, citations	Large distributed knowledge infrastructure
Workspace	Project creation and scoped files/context	Full IDE replacement
Tools	File read/search and controlled basic execution	Unrestricted computer control
Coding	Code explanation and controlled assistance	Fully autonomous software factory
Memory	Conversation/project memory	Unlimited autonomous memory
Security	Permissions and local data boundaries	Enterprise identity federation
Offline	Core local operation	Cloud sync as a requirement

MVP Success Criteria

A user can start NEXUS, detect local Ollama models, chat, switch models, route a task, load local documents, retrieve grounded context with citations, create a project workspace and safely use at least one controlled tool.

MVP Boundary Rule

Anything that does not strengthen the local AI + orchestration + RAG + controlled-agent loop is deferred unless required for security, reliability or usability.

MVP Demonstrations

Conversation → general model; coding request → coding model; complex reasoning → reasoning model; local document question → cited answer; project task → workspace inspection → controlled tool → result.

N04 — User Personas & Use Cases

NEXUS is designed primarily for technical users who want private local AI capabilities while remaining approachable for general knowledge work.

Persona	Need	NEXUS Value
Developer	Code, debug, inspect projects	Coding model + workspace + tools.
Student/Researcher	Study and research privately	Local RAG + citations + summaries.
AI/ML Builder	Experiment with models	Model registry + benchmarking + routing.
Power User	One AI workspace for many tasks	Unified orchestration and memory.
Privacy-conscious user	Keep data on device	Offline-first local inference and storage.

Primary Use Cases

UC-01 local chat; UC-02 automatic model selection; UC-03 model comparison; UC-04 private document Q&A; with citations; UC-05 project workspace; UC-06 controlled coding agent; UC-07 answer/output verification; UC-08 project memory; UC-09 offline operation.

Representative Journey

Install → runtime check → Ollama connection → model discovery → routing → chat → create project → add knowledge → ask grounded question → inspect citations → invoke controlled tool → review result.

N05 — Feature Specification Document

Features are organized by capability and staged so NEXUS can grow without making the MVP unnecessarily complex.

Feature	Core Behavior	Phase
F01 Model Manager	Discover, inspect, enable/disable local models.	MVP
F02 Unified Chat	Chat, streaming, history and conversation controls.	MVP
F03 Model Router	Classify request and select model.	MVP
F04 Model Benchmark	Run repeatable benchmark prompts and compare results.	Advanced
F05 Knowledge Base	Import, organize and index local documents.	MVP
F06 Hybrid Retrieval	Combine semantic and lexical retrieval.	Advanced
F07 Reranking	Improve top-k retrieval relevance.	Advanced
F08 Citations	Map answer claims to retrieved sources.	MVP
F09 Project Workspace	Project-scoped files, context and memory.	MVP
F10 Tool Registry	Register tools with schemas and permissions.	MVP
F11 File Agent	Search/read/write approved project files.	MVP
F12 Terminal Agent	Execute approved commands with restrictions.	Advanced
F13 Coding Agent	Plan, edit, test and iterate in workspace.	Advanced
F14 Agent Planner	Break complex requests into steps.	Advanced
F15 Verifier	Review evidence, outputs and tool results.	Advanced
F16 Memory	Store/retrieve approved conversation/project context.	Advanced
F17 Model Fallback	Recover from unavailable/failed model.	Advanced
F18 System Monitor	Model status, latency, resources and task state.	Advanced
F19 Privacy Center	Data storage, permissions and offline status.	MVP
F20 Web/Cloud Connectors	Optional online models/search/integrations.	Future

Feature Design Rules

Every feature declares its model, tools, permissions, data, failure behavior and observability. Agentic features require explicit stop/cancel behavior.

Safety Rule

NEXUS must never silently escalate tool permissions. File and command execution require explicit policy and controlled scope.

N06 — Product Roadmap

The roadmap evolves NEXUS from a reliable local AI client into a multi-model agentic AI workstation.

Stage	Name	Major Capabilities	Exit Goal
R0	Foundation	Ollama connection, app shell, model discovery, local config.	NEXUS can run locally.

Stage	Name	Major Capabilities	Exit Goal
R1	Local AI Workspace	Chat, streaming, conversations, manual model selection.	Useful local AI chat.
R2	Intelligent Routing	Task classification, routing, fallback and model status.	NEXUS chooses models.
R3	Local RAG	Document ingestion, embeddings, vector search, citations.	Private knowledge assistant.
R4	Agent Tools	Tool registry, file tools, permissions, task execution.	Controlled AI actions.
R5	Coding Agent	Workspace analysis, edits, tests and iteration.	Codex-like local workflow.
R6	Memory & Verification	Project memory, answer verification, evidence checks.	Reliable continuity.
R7	Multi-Agent	Planner, specialist agents, reviewer/verifier.	Complex orchestration.
R8	Evaluation	Model benchmarks, RAG evaluation, agent metrics.	Measurable AI quality.
R9	Productization	Installer, updates, recovery, polished desktop UX.	Shareable local product.
R10	Ecosystem	Optional web, GitHub, cloud models and integrations.	Extensible platform.

Roadmap Philosophy

Build one reliable vertical slice at a time. Do not introduce autonomous agents before model, tool and RAG foundations are observable and permission-controlled.

Future Direction

NEXUS can later support multimodal models, knowledge graphs, browser research, Git integrations, advanced code execution and optional cloud routing.

Product Architecture at a Glance

NEXUS is organized into six conceptual layers.

Layer	Responsibility
Experience	Chat, projects, knowledge, agents, model center and settings.
Orchestration	Routing, planning, context assembly, task state and verification.
Intelligence	Local LLMs, embeddings, rerankers and future multimodal models.
Knowledge	Documents, vector search, retrieval, citations and memory.
Tools	Filesystem, terminal, code execution and future integrations.
Runtime	Ollama, local storage, database, application services and OS resources.

EXPERIENCE → ORCHESTRATION → INTELLIGENCE → KNOWLEDGE / TOOLS → LOCAL RUNTIME

Product Security & Privacy Principles

1. Local-first by default. 2. No secret transmission without explicit user action. 3. Tool access is permissioned. 4. Project workspaces are scoped. 5. Sensitive data is excluded from logs by default. 6. Destructive operations require stronger confirmation. 7. Users can inspect model/tool usage. 8. Offline mode must be testable.

Product Metrics

Track successful model connection rate, chat success rate, routing accuracy, RAG retrieval quality, citation coverage, task completion, tool failure rate, response latency, model load time, resource usage and crash-free sessions.

Acceptance Gate

NEXUS can leave product-foundation phase when MVP scope, user journey, model roles, permissions, architecture boundaries and roadmap are approved. Technical implementation documents then define exact runtime and component choices.

Next Documentation Block

N07 Technical Requirements → N08 System Architecture → N09 Component Architecture → N10 Data Flow → N11 Local AI Runtime → N12 Agent Orchestration → N13 Tool Execution → N14 RAG → N15 Memory → N16 Security & Permissions.

Document status: NEXUS AI Product Design Foundation N01–N06 v0.1.